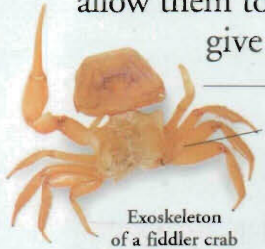


ARTHROPODS

A



MORE THAN ONE MILLION species of arthropod exist, making them the largest group in the animal kingdom. They live in almost all habitats, from mountain tops to the ocean depths. Arthropods are invertebrates – animals without backbones. They come in many shapes and sizes, from tiny mites to large crabs. Their bodies are divided into segments, and they have distinct heads with antennae or eyes. Rigid exoskeletons encase their bodies, but flexible leg joints allow them to move around, and give them their name.



Exoskeleton of a fiddler crab

Exoskeletons are made mainly of the substance chitin.

Exoskeleton

The exoskeleton of an arthropod is a tough outer layer covering the entire body, including the eyes, antennae, and legs. It protects and supports the muscles and soft organs within the body and helps to retain moisture.



Nymph on twig

Moulting and growth

Exoskeletons are fixed in size. In order to grow, an arthropod must shed, or moult, this rigid layer. Its body then rapidly expands before a new exoskeleton hardens in place of the old one. Moulting is part of a process called "incomplete metamorphosis". This is where the young, called nymphs, emerge from eggs looking like tiny adults. They moult many times before reaching adult size.

In "complete metamorphosis", the animal changes form as well as size.

Moulting

1 An emerging adult grasshopper has cracked open its old exoskeleton and is starting to wriggle its body free, headfirst. Before this final moult, the nymph will already have been through four previous moults.



Adult is almost free of the nymph's skin.

2 The adult has pulled its legs and most of its body out of the old skin. It is already expanding in size, now that it is free from its confines.



Old, empty exoskeleton

Adult waits as blood pumps into its wings before it flies away.

3 Moulting is now complete. The adult rests while its new exoskeleton hardens and its wings unfurl. Its old exoskeleton, now empty and brittle, still clings to the twig.

Feeding

Arthropods feed on all kinds of plant and animal matter, both living and dead. Some arthropods, such as praying mantises, have pincers to gather food; others use their front legs. Many have cutting and chewing mouthparts, while those that feed on fluids, such as true bugs, have mouths modified for sucking. Small aquatic arthropods eat by filtering food particles from water.



Field chafer beetle

Herbivores

Some arthropods, such as chafer beetles, eat only plant matter. Adults feed on stems, leaves, and buds, while larvae eat plant roots.

Carnivores

Many arthropods feed on other animals. Garden spiders, for example, feed mainly on insects. Some meat eaters also eat dead animals and are called scavengers. Sand crabs scavenge on dead birds and other debris found on the beach and seabed.

Web spun around wasp



Garden spider feeding on a wasp

Defence

As arthropods are generally small in size, they are the target for a great many predators. Their hard exoskeleton, which acts as a tiny suit of armour, provides the first line of defence. Some arthropods, such as pill millipedes, take a passive form of defence and roll up into a ball if danger threatens. Other arthropods have special protective weapons; including stings and pincers. Many ant species have glands on their abdomens from which they secrete formic acid to drive off enemies.

Stings and pincers

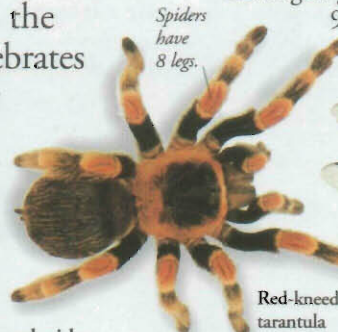
Some arthropods have pincers and stings which they use to defend themselves against attackers. Scorpions also use their large pincers to catch animals. They then use their venom-filled stings to paralyze their prey.



Fat-tailed scorpion

Types of arthropod

Arthropods vary in size, from minute creatures a fraction of a millimetre long to outsized sea dwellers several kilograms in weight. There are four main types of arthropod – insects, arachnids, crustaceans, and myriapods. Insects are the largest group, accounting for almost 90 per cent of all arthropods.



Red-kneed tarantula

Spiders have 8 legs.



Large compound eye helps it to catch prey in flight.

Broad-bodied chaser dragonfly

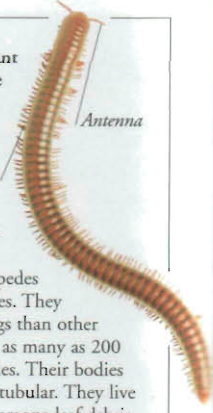
Insects

Insects are the most diverse group of arthropods. They live in all kinds of land and freshwater habitats. All adult insects have six legs, and most have wings – they are the only arthropods that can fly.

Delicate wings

Asian giant millipede

Two pairs of legs on each body segment



Antenna

Myriapods

Myriapods include millipedes and centipedes. They have more legs than other arthropods – as many as 200 in some species. Their bodies are long and tubular. They live in the soil or among leaf debris.

European lobster



Hard exoskeleton

Crustaceans

Crustaceans include crabs, shrimps, and lobsters. Most live in the sea or in freshwater and have five pairs of legs. Lobsters and crabs have very thick exoskeletons and some grow extremely large.

Reproduction

Breeding habits are very diverse among arthropods. Fertilization may take place inside or outside the female's body. Normally, eggs are laid; some are guarded, others are hidden and left alone. The young of some arthropods, such as garden spiders, are tiny versions of adults called nymphs; others start life as larvae and look different from the adults.



Cluster of young garden spiders

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